

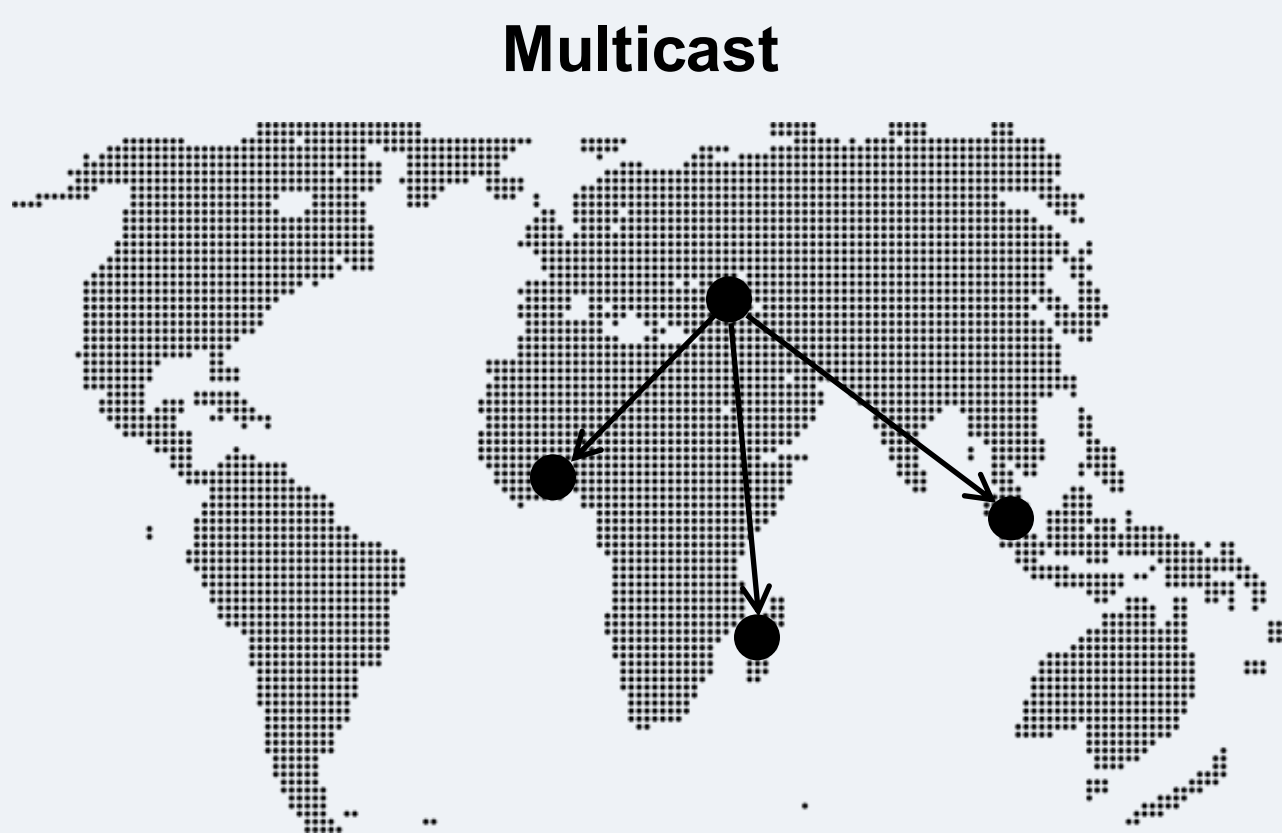
Quantum Multicast Communication for the Quantum Internet

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Quantum Internet

- Classical networks can copy data to send to multiple receivers – “Multicast”
- No cloning theorem makes quantum multicast communication more complex
- Useful for: distributed QC, multiparty cryptograph, clock synchronization
- How could we do this with existing network?

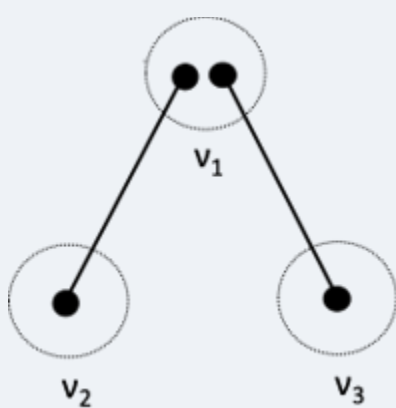


Multicast Methods

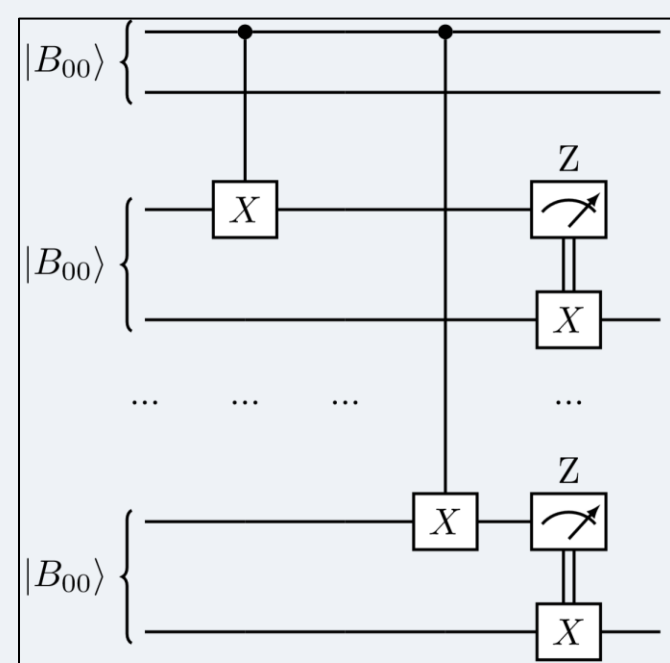
- Two methods: Multipartite and Multiple-bipartite
- Multipartite: Share GHZ state
 - Lower memory requirement
- Multiple-bipartite: Share Bell pairs, LOCC
 - More resistant to noise

Bell Pairs

$$|B_{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

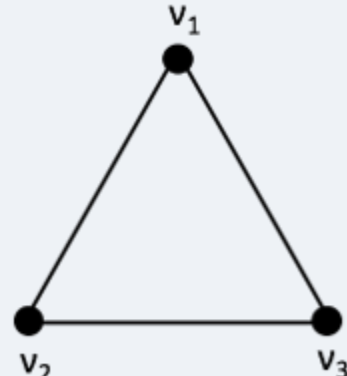


LOCC



GHZ States

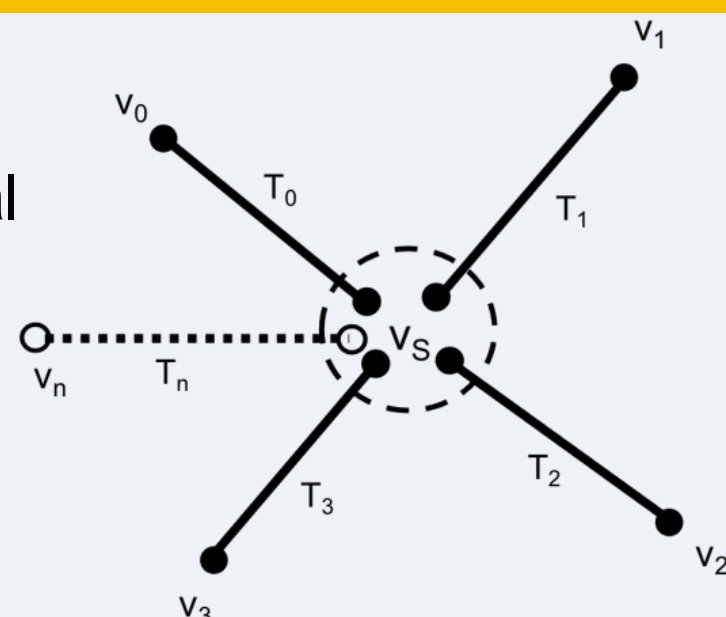
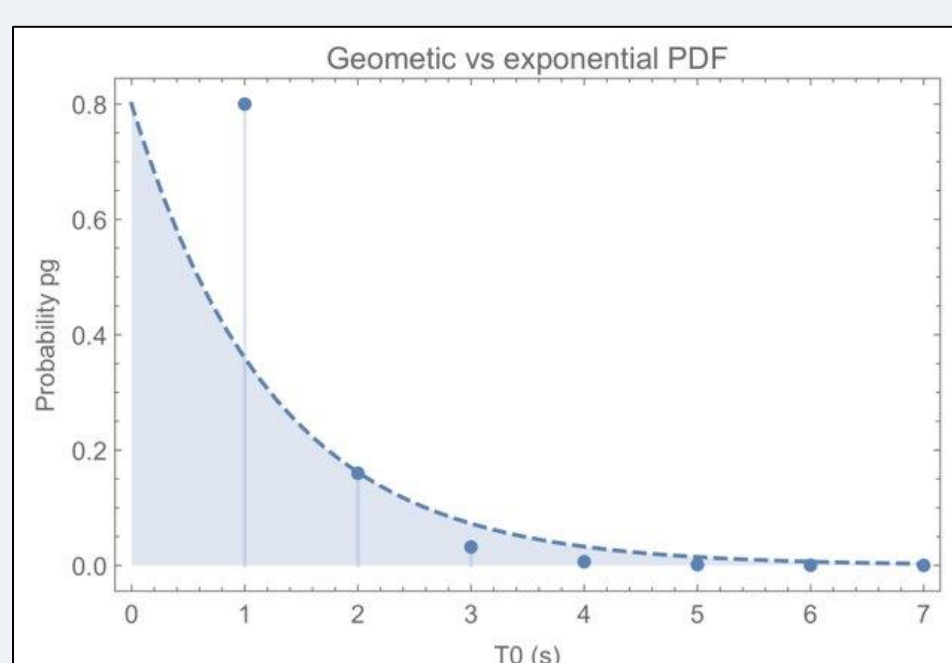
$$|GHZ_N\rangle = \frac{|0\rangle^{\otimes N} + |1\rangle^{\otimes N}}{\sqrt{2}}$$



Link Generation Rate

Analytic Approach

- Approximate probability with exponential $P(T_0 = t) = p_g e^{-p_g t}$



- Input loss & fiber loss

$$p_g = p_l e^{-\frac{d}{L_0}}$$

Network Simulation

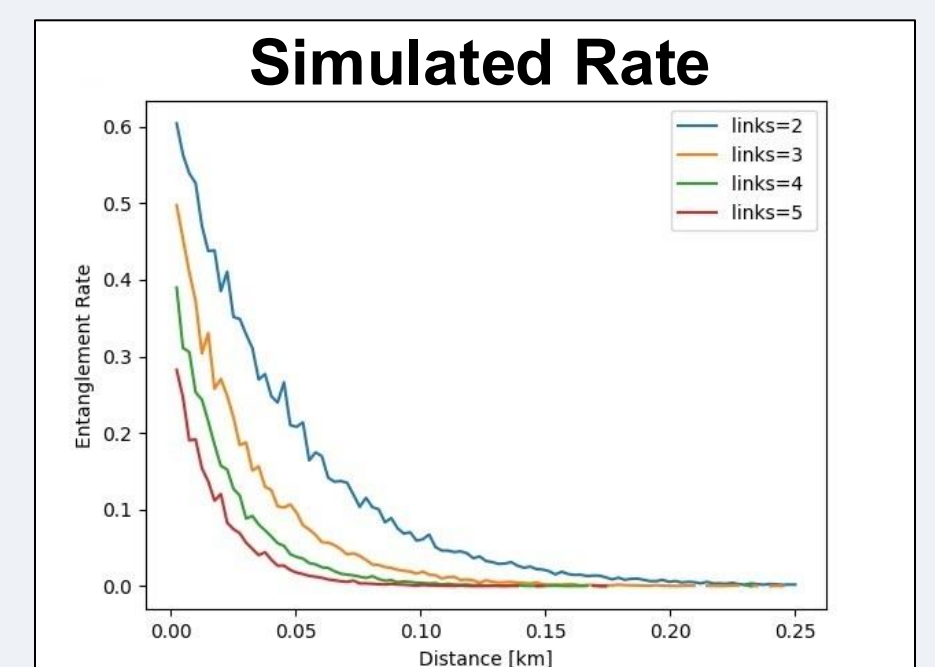
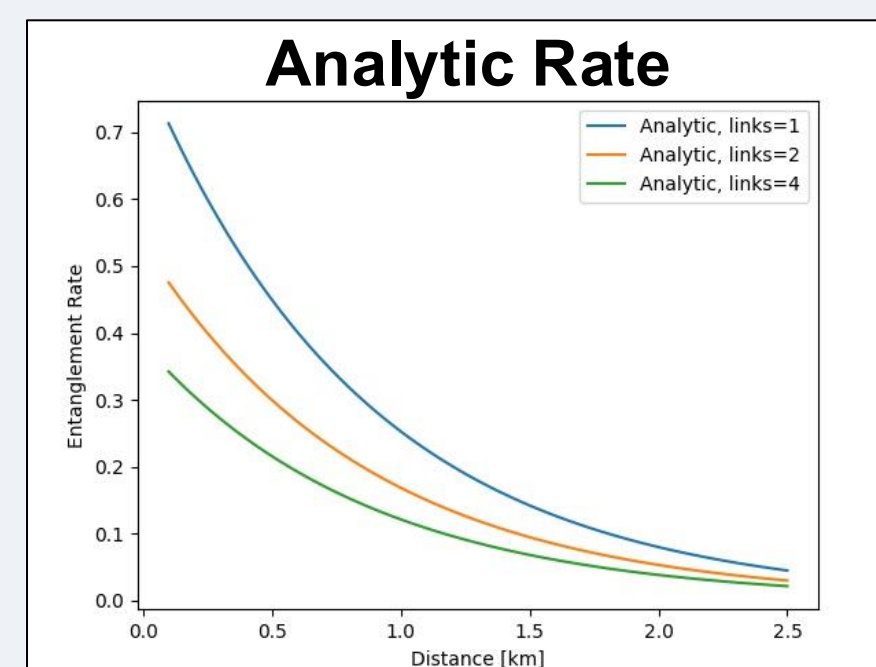
- Should be fast, flexible & realistic
- NETSQUID
- Assume error free LOCC
- Pauli noise model for fibers $P = 1 - \exp(\frac{-t\gamma}{10^{-9}})$



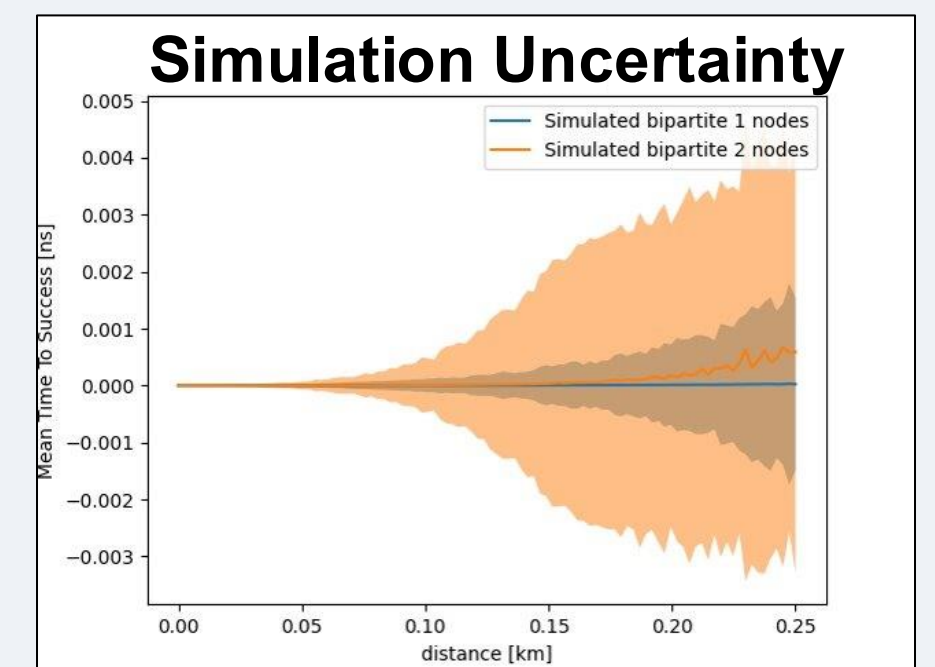
GitHub Repo



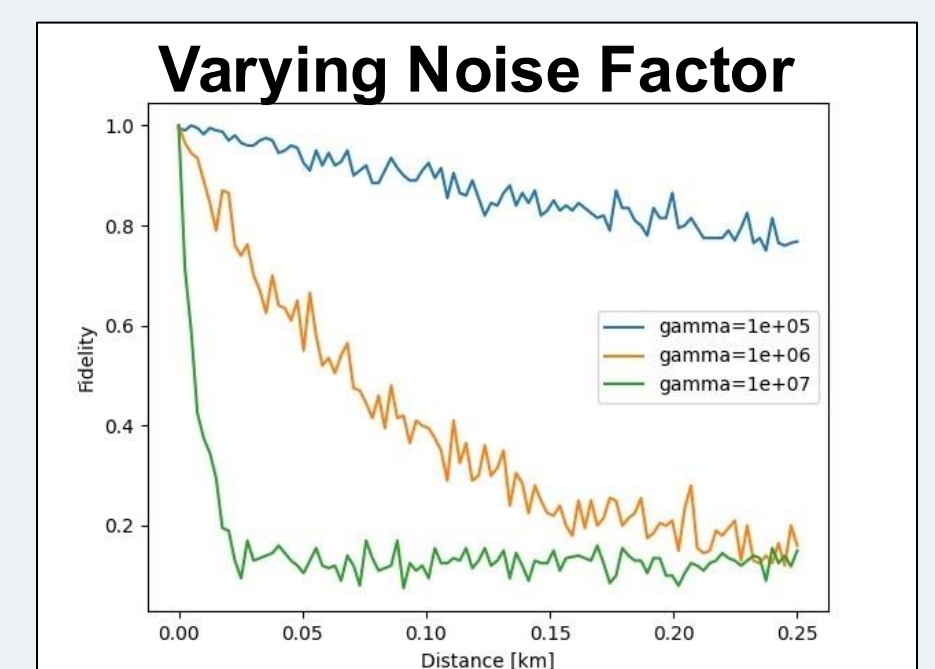
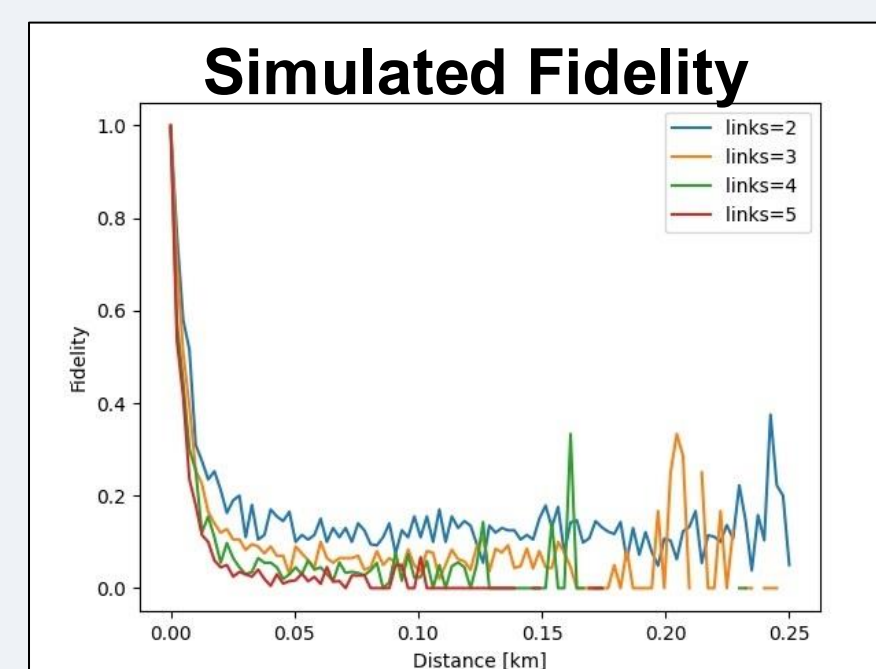
Results



- The analytic rate of successful entanglement is much higher than the simulation rate – approximating as an exponential works best for unlikely transmission which is time consuming to simulate.



- Fidelity falls rapidly, to the lower limit of 2^{-n} where n is the number of connections. Fiber noise needs to be very low to allow for useful communication.



Future Work

- Analytic characterization of high probability transmission.
- LOCC noise & transmission time.
- Repeaters and complex networks.
- Simulate entanglement distillation to find rate at which states can be prepared for communication.

